

Lecture Notes 12

Summary and Review

Intermediate Macroeconomics, EC2211

Martin Flodén

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Department of Economics, Stockholm University

Lecture 1: Measuring the macroeconomy

Measuring the macroeconomy

- Measurement
 - The national income identity; three approaches to accounting for GDP
 - Distinguish between nominal variables ("current prices") and real variables ("volumes")
 - Price indices and PPP conversion
- Growth facts

Note how the national income identity has been used repeatedly in the course!

Lectures 2-3: Economic growth and the Solow model

The production function and growth accounting

- The production function
 - The Cobb Douglas production function is used widely in macroeconomic models

$$Y = AK^{\alpha}L^{1-\alpha}$$

- Constant returns to scale, diminishing marginal products
 - α as the capital share, $1 - \alpha$ as the labor share
- Growth accounting
 - If $y = Ak^{\alpha}$ then $g_y \approx g_A + \alpha g_k$, etc
 - g_A is calculated as the residual when the other growth rates and parameters are "known"

Growth accounting and development accounting

Growth accounting

- Suppose Y increases from Y_1 to Y_2 from period 1 to period 2
- How much of the increase is explained by A , K and L , respectively?
 - Or use more details on production factors
 - ... for example, by accounting for hours and employment behind labor input
 - Or by including human capital as a production factor

Development accounting:

- Same as growth accounting, but explain differences between countries
- Example: How much of the difference in income per person between country A and country B (Y_A/L_A vs Y_B/L_B) is explained by A and K/L , respectively?

The Solow model

- Production function: $Y_t = A_t K_t^\alpha L_t^{1-\alpha}$
- Development of capital stock: $K_{t+1} = K_t + I_t - \delta K_t$
- Closed economy: investment = savings = constant fraction s of output
- Convergence to a steady state:
 - Capital stock increases if $sY_t > \delta K_t$
 - Capital stock decreases if $sY_t < \delta K_t$
 - Steady state with constant capital stock: $sY_t = \delta K_t$
- Understand how the analysis in the Solow model is affected by population growth and human capital in the production function
 - Reformulate the model in per-capita terms
 - Understand how the balanced growth path is determined (the steady state for the per capita variables)

The Solow model

- More savings (and thus investment) result in higher production
 - ... but not sustained growth
 - ... and not necessarily higher consumption (recall the golden rule result)
- The importance of decreasing marginal product of capital
- Understand how changes in, for example, the depreciation rate, the savings rate, the population growth rate, productivity, amount of capital or the number of workers affect the economy in the short and long run

The Solow model

- (Absolute) convergence:
 - Suppose two countries are "similar"
 - ... i.e. similar productivity, savings rate, population growth rate, etc
 - Then they should both converge towards same steady state
 - Countries far below that steady state should grow fast ("catch up")
- Conditional convergence:
 - Understand how the above result is affected if countries are different
 - Countries then have different steady states
 - ... and distance to respective steady state affects growth rates

Finite natural resources

- Version of the Solow model where:
 - the stock of one production factor ("oil") falls over time as it is used in production and cannot be rebuilt
 - or the stock of one production factor ("land") is fixed
- A drag on economic growth
- But positive long-run growth is still possible if technological progress is sufficiently fast

The Malthusian model

- Land is important production factor, but in fixed supply
- Diminishing marginal product of labor (because of crowding on the land)
- Population growth is endogenous, depends on income
- Understand why income per capita cannot grow in the long run even if there is technological progress
(Or in level terms: why income per capita does not increase in the long run even if technology improves)

Lecture 4: TFP and long-term economic growth

The Romer model

- Endogenous TFP: research and development $\rightarrow$ ideas, higher productivity
- Capital and labor are rivalry production factors, diminishing marginal products
- Ideas are not rivalry: they can be used by everyone
- Understand how to find the balanced growth path
 - How is the economy affected in the short and long run by, for example, a higher research productivity, having more workers in the R&D sector, more people in the economy or a more rapid population growth rate?

Lecture 5: Labor supply, unemployment, consumption and investment

The labor-leisure choice

- Static, one-period, model
- Household chooses how much to work and how much leisure to enjoy and how much to consume
- Optimization problem:
 - Household preferences, utility function
 - Budget constraint: income, and therefore consumption, depends on hours worked
- Understand how to formulate the relevant budget constraint and how to solve the optimization problem
- Understand how changes in, for example, the wage rate or the income tax rate affect the labor-leisure choice
 - Income and substitution effects

- Definitions of unemployment rate, employment rate, labor-force participation rate
- Other concepts: natural rate of unemployment, frictional unemployment, structural unemployment, cyclical unemployment
- The bath-tub model of unemployment
 - Flows on the labor market
 - How job finding and separation rates affect the steady-state unemployment level

The consumption-saving choice

- Two-period model
- Household chooses how to allocate consumption between the two periods
- Optimization problem:
 - Household preferences, utility function
 - Budget constraint: savings and the interest rate is the link between the two time periods
- Understand how to formulate the relevant budget constraint and how to solve the optimization problem
- Understand how changes in, for example, the interest rate or income affect the consumption-saving choice
 - Income and substitution effects
 - Consumption smoothing and the permanent income hypothesis

- Return on financial capital (r) and return on physical capital ($MPK - \delta$)
- The *user cost of capital*
- Investment choice negatively related to level of interest rate

Lectures 6-7: The short run

The IS curve

- Downward-sloping relation between interest rate and economic activity (output gap)
- National income account identity: $Y = C + G + I(r) + NX$
- Negative relation because:
 - $I(r)$ negatively related to r
- IS curve: $\tilde{Y}_t = \alpha - \beta(r_t - \bar{r})$
- IS curve shifts out if:
 - α or $\bar{r}$ increases
 - I.e. if something happens that raises "demand" ($C + G + I + NX$) for a given interest rate

The IS curve and multipliers

- National income account identity: $Y = C(Y - T) + G + I(r) + NX$
- Positive relation between consumption, C , and disposable income, $Y - T$?
 - The *marginal propensity to consume*, MPC
 - The MPC out of temporary fluctuations in disposable income should be low according to permanent income hypothesis
 - But if borrowing constraints, or just "Keynesian behavior", MPC can be larger
- If C responds to temporary increases in $Y - T$:
 - There are multiplier effects
 - ... positive demand shocks raise income which leads to higher consumption demand (an additional positive demand shock), etc
- In a graph, the IS curve is still the same, just flatter

Monetary policy and the MP line

- The Fisher equation: $i_t = r_t + \pi_t^e$
- The central bank sets the nominal interest rate
- Prices are sticky in the short run, inflation expectations assumed to be exogenous
- The central bank then also sets the real interest rate, the MP line
- MP line shifts up if the central bank raises the policy rate
 - But also if it holds the nominal interest rate constant when inflation expectations fall (see Fisher equation)

The Phillips curve

- Positive relation between economic activity (the output gap) and inflation's deviation from expected inflation: $\pi_t = \pi_t^e + \nu \tilde{Y}_t$
(or negative relation between unemployment and inflation)
- Shifts up if inflation expectations increase or if positive cost shock
- Movement along the curve if economic activity changes

The IS-MP-PC framework

- Understand how the framework is affected by "shocks":
 - Demand shocks (things that shift the IS or MP curve)
 - Supply shocks (price shocks)
 - Economic policy: changes in i , G and T
 - Inflation expectations

The AS-AD model

- A monetary policy rule: $r_t = \bar{r} + \mu(\pi_t - \pi^*)$
How the central bank typically adjusts its policy rate when inflation changes
- AD curve combines MP rule and IS curve: $\tilde{Y}_t = \alpha - \beta\mu(\pi_t - \pi^*)$
 - Shifts out when IS curve shifts out¹
 - Also shifts if the MP rule changes (a new inflation target)
- AS curve is basically Phillips curve: $\pi_t = \pi_{t-1} + \nu\tilde{Y}_t + \sigma_t$
 - Adaptive inflation expectations: $\pi_t^e = \pi_{t-1}$
 - Cost-push shocks (price shocks), σ

¹The exception is if $\bar{r}$ changes, because this MP rule then automatically adjusts

The Taylor rule

- How central banks typically have responded to changes in inflation and economic activity
- The **Taylor principle**:
 - Suppose that inflation (and inflation expectations) increase by one unit
 - Recall Fisher equation, $r_t = i_t - \pi_t^e$
 - To raise the real interest rate, the central bank must raise the nominal interest rate *more* than the increase in inflation (if $\pi_t^e = \pi_{t-1}$)

Lecture 8: Money and monetary policy

Money and banking

- Banks
 - Bank balance sheets
 - Maturity transformation and leverage
 - Liquidity provision, payment services and money creation
- Bond prices and interest rates
 - Yield curve and term structure of interest rates
 - Forward curve
- Money
 - The functions of money
 - Monetary base and broad money aggregates

The transmission mechanism of monetary policy

- Transmission channels
 - Interest-rate channel
 - Cash-flow channel
 - Banking channel
 - Expectations channel
- Frictions in the transmission
 - Banks under stress
 - Financial markets under stress: risk premia and fire sales
 - Policy rate at effective lower bound
 - Loss of central bank credibility, inflation expectations de-anchoring from inflation target

Central bank's toolbox to address frictions

- Lender of last resort
 - Lend against collateral
- Market maker of last resort
 - Step in as buyer on financial markets
- Communication, forward guidance
 - Commit to holding policy rate low for long
 - Commit to keeping risk premia low through purchases
 - ...
- Other possible measures (not necessarily by central banks):
 - Bail-out or nationalization
 - Deposit insurance
 - Regulation, macro-prudential measures (before crisis)

Time-inconsistency problem

- Phillips curve
- Policymaker that wants inflation at target and unemployment below the "natural" rate
- Risk that policymaker tries to exploit Phillips curve
 - Would result in inflation bias: inflation above target but unemployment not below the natural rate
- Solution: set up institutions that generate longer time-horizons in central banking and that facilitate accumulation of credibility and reputation

Lecture 9: Government debt and fiscal policy

Debt dynamics and fiscal sustainability

- The total and primary surplus
- Which primary surplus is needed to stabilize the debt ratio?
- A target for the total surplus and effects on the long-run debt ratio
- The importance for debt dynamics of the interest rate (on government debt) relative to the growth rate in the economy
- Expectations, risk premia and vicious circles
- Debt, default and inflation

- Does a reduction of taxation today make households "richer"?
 - Forward-looking households understand that someone will eventually have to pay for the tax cut
 - If RE holds: a tax cut does not affect household consumption
- Assumptions for RE to hold

- Active fiscal policy and automatic stabilizers
- Effects of changes in G and T on the IS- and AD-curves
 - Effect of T depends on if, or to what extent, ricardian equivalence holds
 - Effect of G may similarly depend on how persistent the change is and how households think G will be funded
- Coordination of fiscal and monetary policy?

Lectures 10-11: International finance

- Nominal and real exchange rates
- The law of one price, purchasing power parity and the Balassa-Samuelson effect
- Interest rate parity

Exchange rates in the Keynesian framework

- Net exports depend on the exchange rate
- And exchange rate depends on interest rate (through interest rate parity)
- IS curve in open economy with flexible exchange rate:
 - Exchange rate channel
 - ... reinforces the interest-rate channel
 - International transmission of monetary policy

Exchange rate regimes

- The trilemma of international finance
- Importance of expectations and credibility
- Currency crisis
- International reserve currency
 - The Triffin dilemma
 - The exorbitant privilege

The balance of payments

- The current account balance
- International capital flows
 - $CA = S - I = NX + \text{net factor income from abroad}$
- Dynamics of the international investment position
- Unwinding of a persistent trade deficit
 - Do tariffs reduce imports and reduce the deficit?
 - Or do they lead to currency offset?

Some final reflections

Malthus → Solow → Romer

Land → Capital → Ideas

Note in particular the roles of populations and technological progress in these economies:

- In the rural economy, land is scarce
 - More people means less land per person
 - Technological progress will enable more food production and more people, not more income per person
- In the industrial economy, decreasing marginal product of capital
 - But more capital can be built if the population increases
 - Technological progress is key for rising income per person
- In the R&D economy
 - More people leads to more ideas, more technological progress and rising income per person

The Great Inflation: Why so high inflation in the 1970s?

- Oil price shocks and other negative supply shocks
- Breakdown of the Bretton-Woods system
 - Last backing by gold was lost
 - Search for a new nominal anchor with fiat currencies
- "Discovery" of the Phillips curve
 - ... which was not well understood
 - ... so the PC shifted up when policymakers tried to exploit a tradeoff between unemployment and inflation
 - The tradeoff also induced political pressure on central banks
- The time inconsistency problem
 - Central banks were not sufficiently independent and lacked other institutional features to build credibility
 - ... so π^e increased even when the PC was better understood
- Mismeasurement of the output gap when productivity growth slowed down

The importance of expectations

We have touched on several episodes where expectations risk becoming self-fulfilling and taking the economy to an undesired equilibrium:

- Great Depression: bank runs
 - Withdraw deposits just because you are afraid that others will
- Global Financial Crisis and pandemic: fire sales
 - Lower bond prices raises funding costs, and makes default more likely, pushing bond prices further down
- European debt crisis: default risk
 - Again, expectations of default leading to higher funding costs and more likely default
- Swedish 1990s crisis: devaluation risk
 - Risk of devaluation forces central bank to raise interest rate, making devaluation more tempting
- Low-inflation episodes: deflation risk
 - If inflation expectations fall too much, the economy may get stuck with the policy rate at its lower bound and a too high real interest rate, resulting in very low inflation

The importance of expectations

These episodes illustrate the importance for economic policy to manage expectations:

- Proactive policy to keep fundamentals away from danger-zones ...
 - Public and private debt at levels that are robust to shocks
 - Inflation and monetary policy in line with formulated objectives
 - Well-capitalized and profitable banks
 - Transparent communication
- ... which in turn requires efficient institutions
 - Fiscal frameworks
 - Macroprudential frameworks
 - Inflation targets and independent central banks
 - Bank regulation and deposit insurance
- Crisis policy
 - Being able and willing to quickly step in
 - Sometimes "the power of words" is enough
 - ... but often a lender or market maker of last resort is needed, or government guarantees